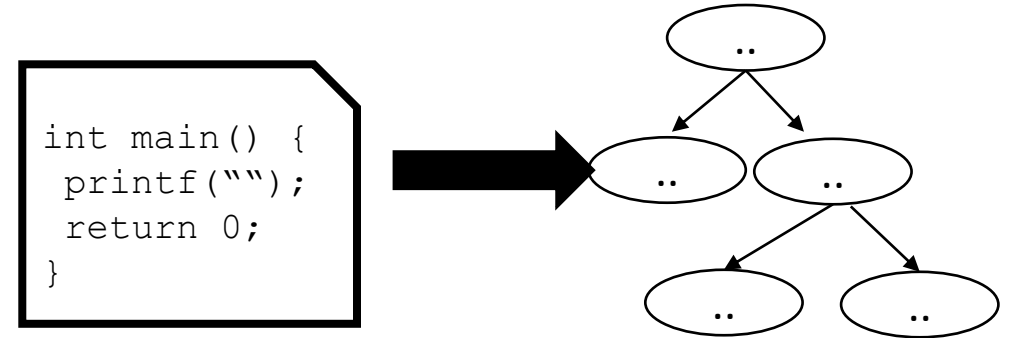


CSE110A: Compilers

Topics:

- *Syntactic Analysis continued*
 - *Derivations*
 - *Parse trees*
 - *Precedence and associativity*



Ambiguous grammars

- What happens when different derivations have different parse trees?

```
1: Statement ::= "if" Expr "then" Statement "else" Statement
2:           |   "if" Expr "then" Statement
3:           |   Assignment
4:           |   ....
```

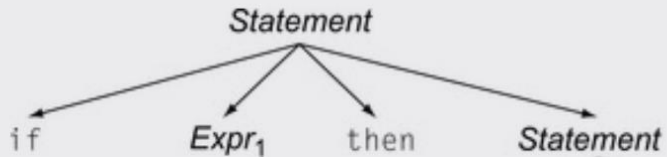
can we derive this string?

`if Expr1 then if Expr2 then Assignment1 else Assignment2`

Ambiguous grammars

```
1: Statement ::= "if" Expr "then" Statement "else" Statement
2:           |  "if" Expr "then" Statement
3:           |  Assignment
4:           |  ....
```

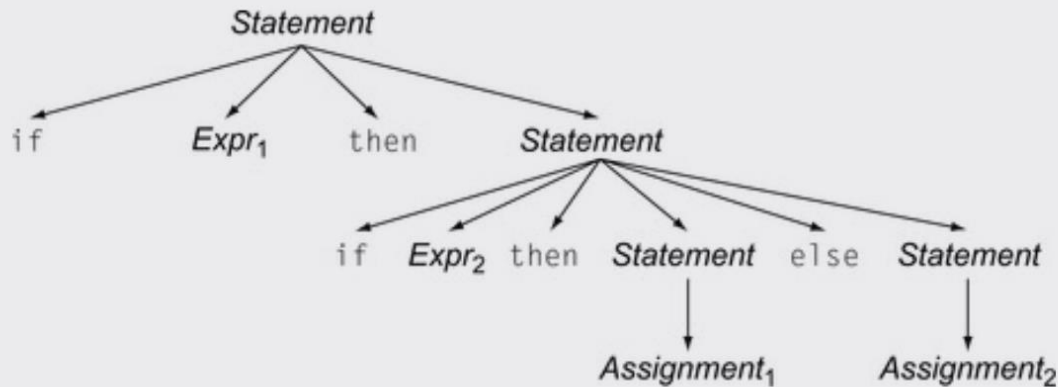
`if Expr1 then if Expr2 then Assignment1 else Assignment2`



Ambiguous grammars

```
1: Statement ::= "if" Expr "then" Statement "else" Statement
2:           |   "if" Expr "then" Statement
3:           |   Assignment
4:           |   ....
```

`if Expr1 then if Expr2 then Assignment1 else Assignment2`

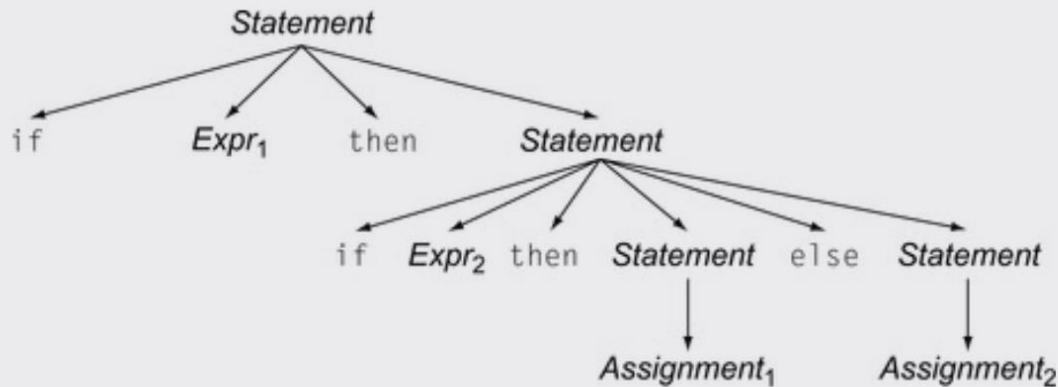


Valid derivation

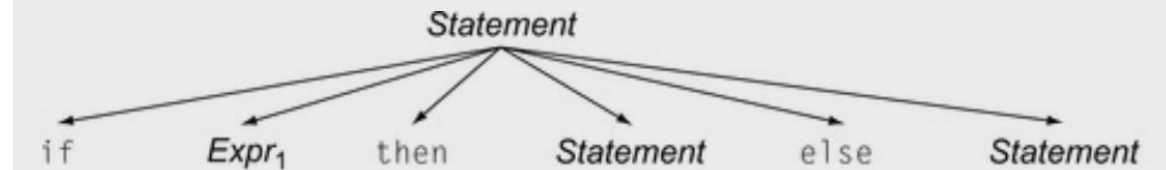
Ambiguous grammars

1: Statement ::= "if" Expr "then" Statement "else" Statement
2: | "if" Expr "then" Statement
3: | Assignment
4: |

if Expr₁ *then* *if* Expr₂ *then* Assignment₁ *else* Assignment₂



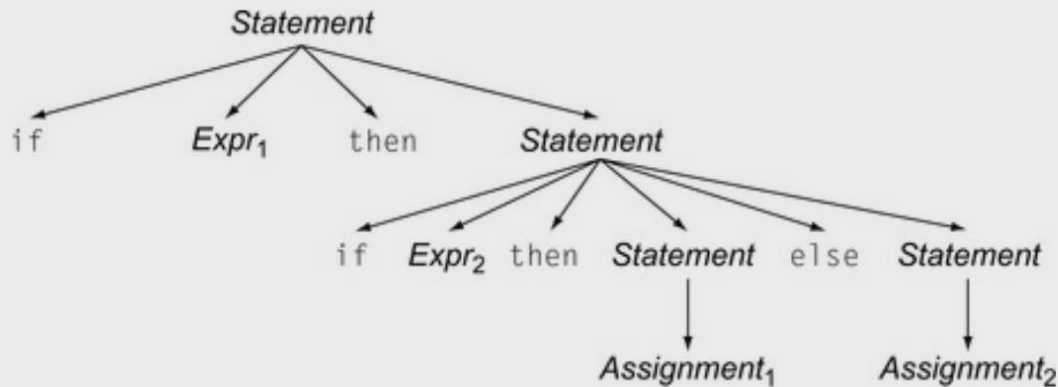
Valid derivation



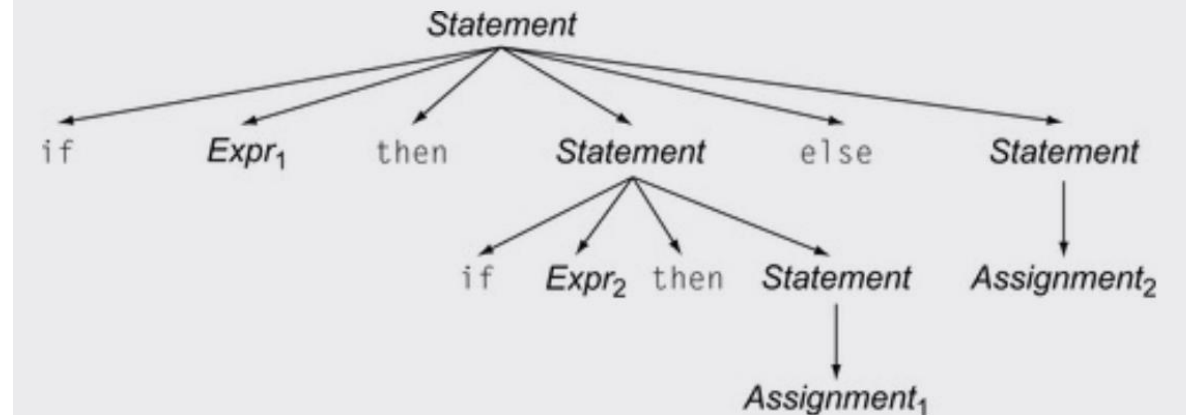
Ambiguous grammars

1: Statement ::= "if" Expr "then" Statement "else" Statement
2: | "if" Expr "then" Statement
3: | Assignment
4: |

`if` Expr₁ `then` `if` Expr₂ `then` Assignment₁ `else` Assignment₂



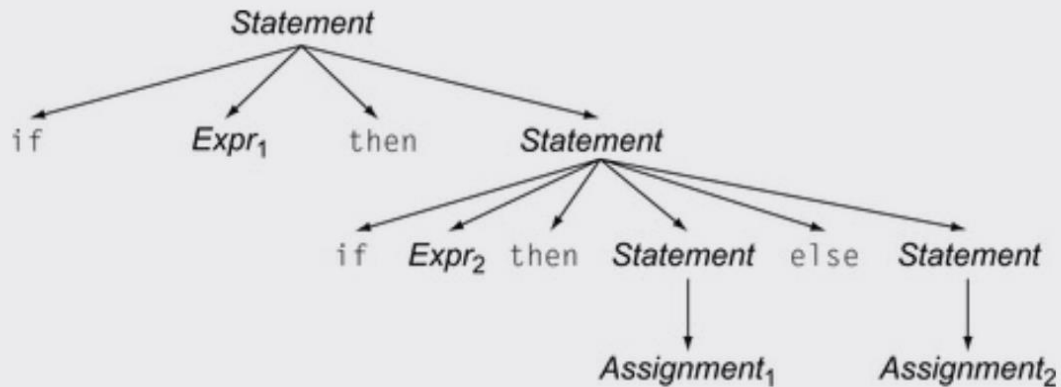
Valid derivation



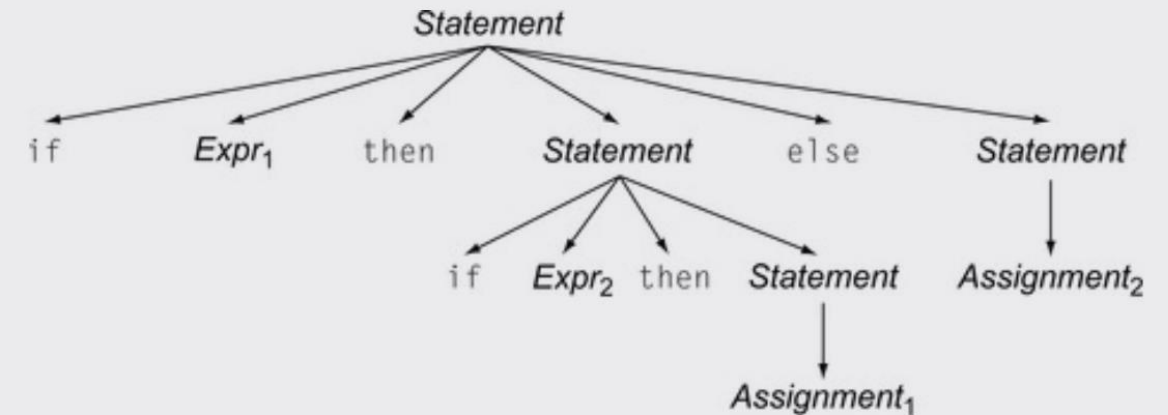
And another valid derivation

Ambiguous grammars

Is this an issue? Don't we only care if a grammar can derive a string?



Valid derivation

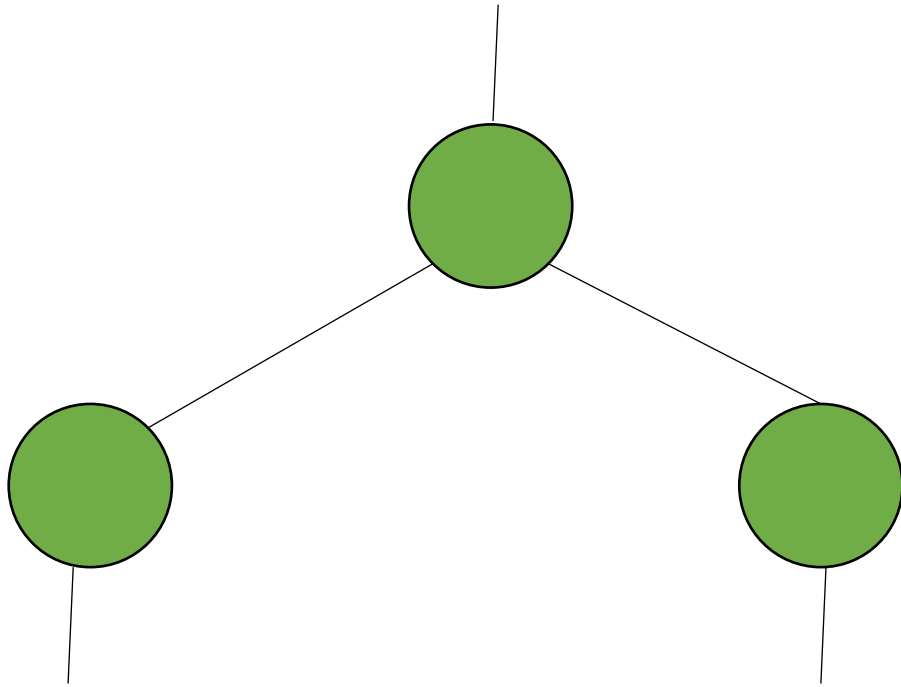


And another valid derivation

Meaning into structure

- We want to start encoding meaning into the parse structure. We will want as much structure as possible as we continue through the compiler
- The structure is that we want evaluation of program to correspond to a post order traversal of the parse tree (also called the natural traversal)

Post order traversal



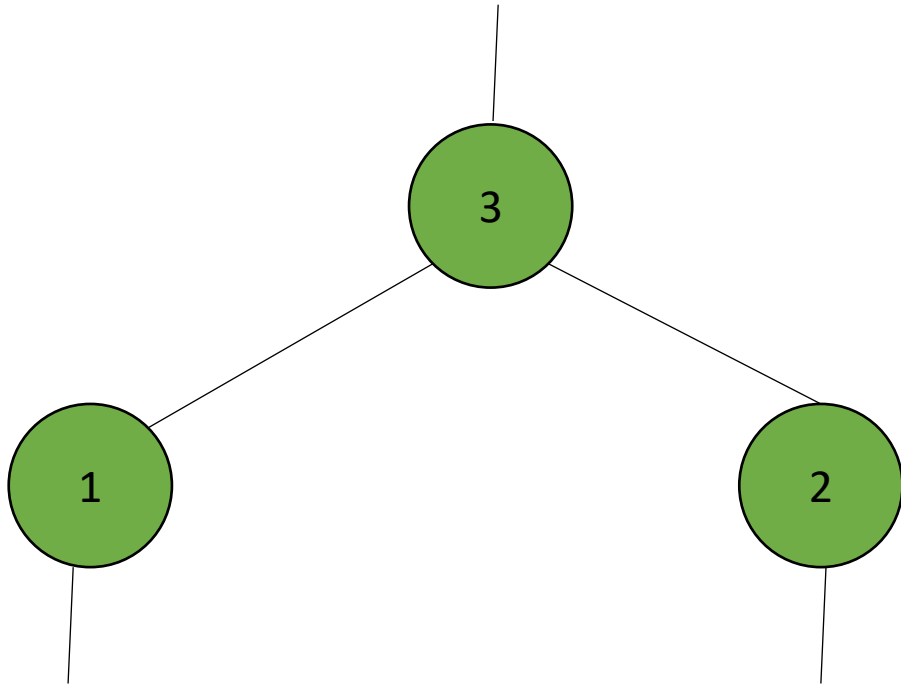
visiting for for different types
of traversals:

pre order?

in order?

post order?

Post order traversal

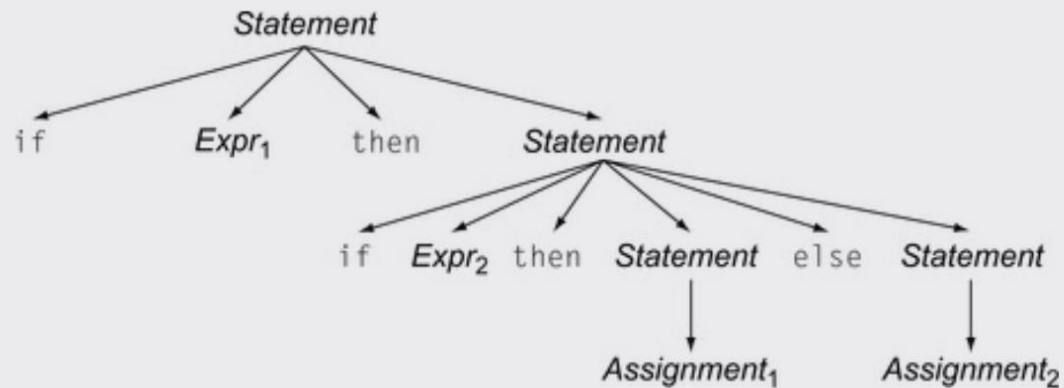


visiting for for different types
of traversals:

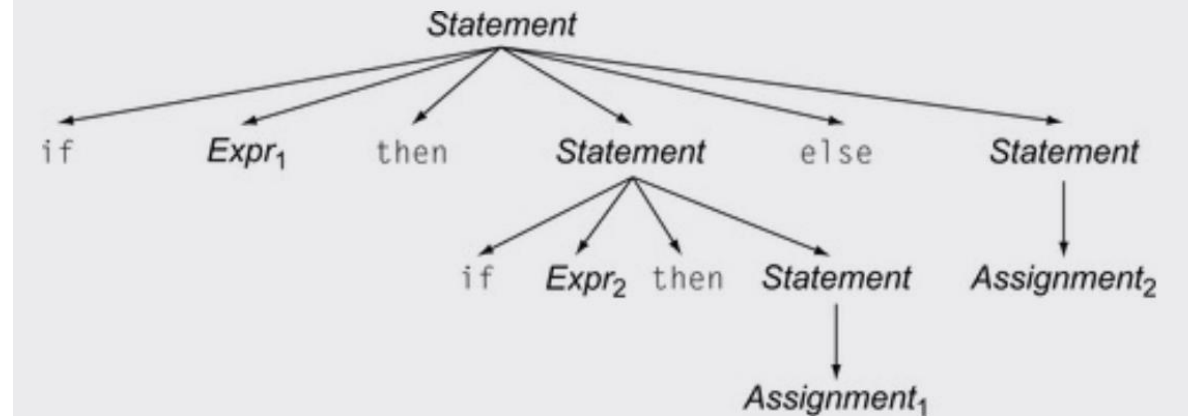
post order

Ambiguous grammars

Encoding meaning into structure can result in very different programs



Valid derivation



Also a valid derivation

Programming language structure

```
int x = 1; //true  
int y = 0; //false  
int check0 = 0;
```

```
if (x)  
if (y)  
    pass();  
else  
    check0 = 1;
```

pop quiz: what is the value of check0 at the end?

Programming language structure

```
x = 1  
y = 0  
check0 = 0
```

```
if (x):  
    if (y):  
        pass  
    else:  
        check0 = 1
```

```
print(check0)
```

How does Python handle this?

Programming language structure

```
x = 1  
y = 0  
check0 = 0
```

```
if (x):  
    if (y):  
        pass  
    else:  
        check0 = 1
```

```
print(check0)
```

How does Python handle this?

```
x = 1  
y = 0  
check0 = 0
```

```
if (x):  
    if (y):  
        pass  
    else:  
        check0 = 1
```

```
print(check0)
```

Invalid syntax, you need to indent, which makes it clear

Ambiguous expressions

- First let's define tokens:

- NUM = "[0-9]+"
- PLUS = "\+"
- TIMES = "*"
- LP = "\("
- RP = "\)"

Let's define a simple expression language

```
Expr ::= NUM
      | Expr PLUS Expr
      | Expr TIMES Expr
      | LP Expr RP
```

Parse trees examples

input: 5

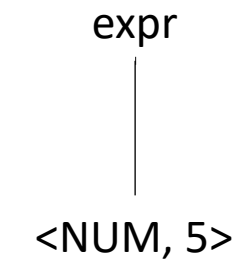
```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

expr
|

Parse trees examples

```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

input: 5



Parse trees examples

input: 5*6

```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

Parse trees examples

input: 5*6

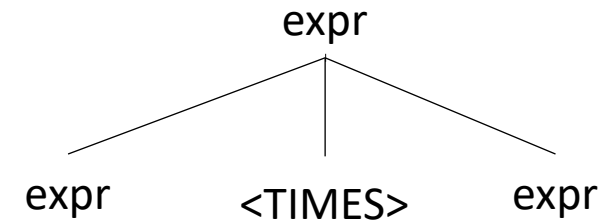
```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

expr

Parse trees examples

```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

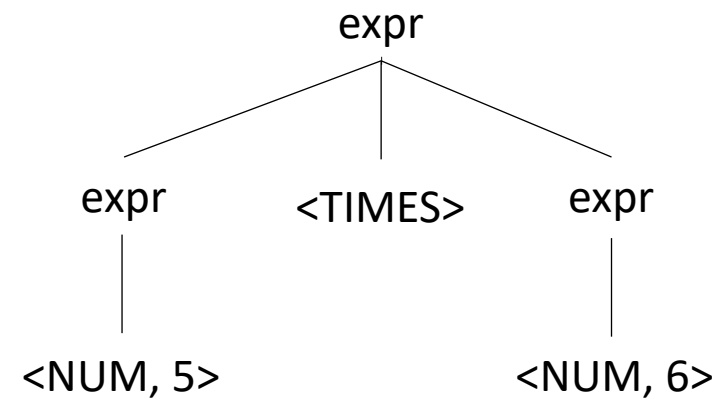
input: 5*6



Parse trees examples

```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

input: 5*6



Parse trees examples

```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

input: 5**6

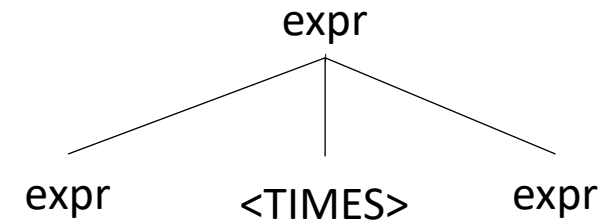
expr
|

What happens
in an error?

Parse trees examples

```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

input: 5**6



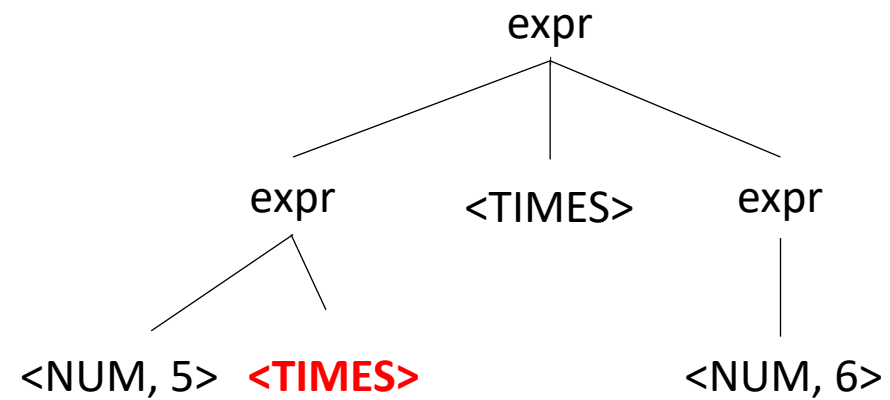
What happens
in an error?

Parse trees examples

```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

input: 5**6

What happens
in an error?



Not possible!

Parse trees examples

input: (1+5)*6

```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

Parse trees examples

input: (1+5)*6

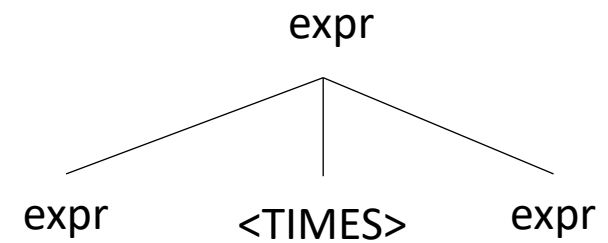
```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

expr

Parse trees examples

input: (1+5)*6

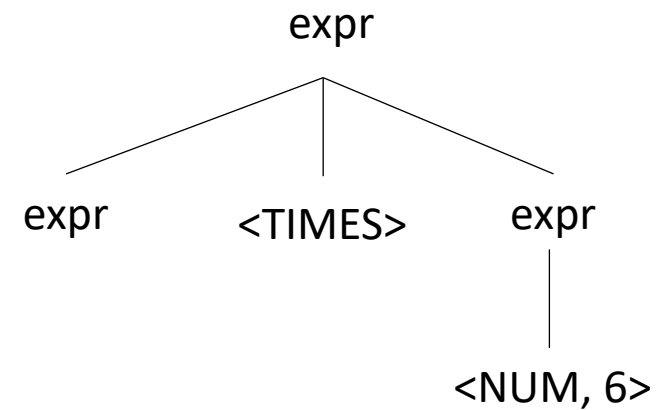
```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```



Parse trees examples

input: (1+5)*6

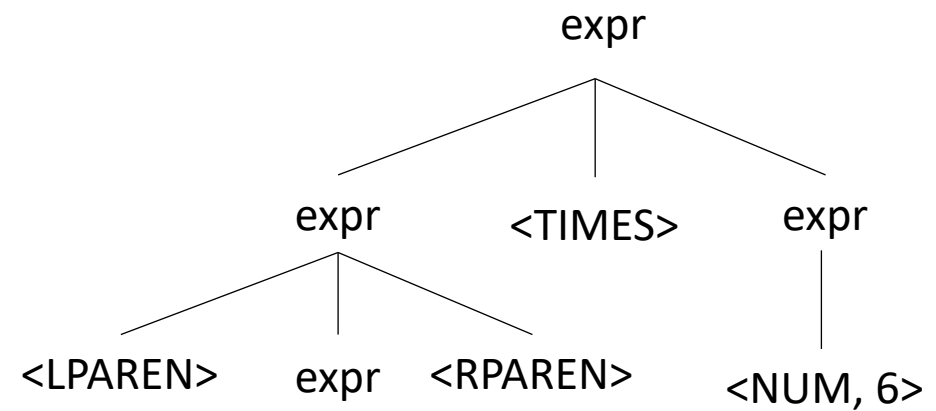
```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```



Parse trees examples

input: (1+5)*6

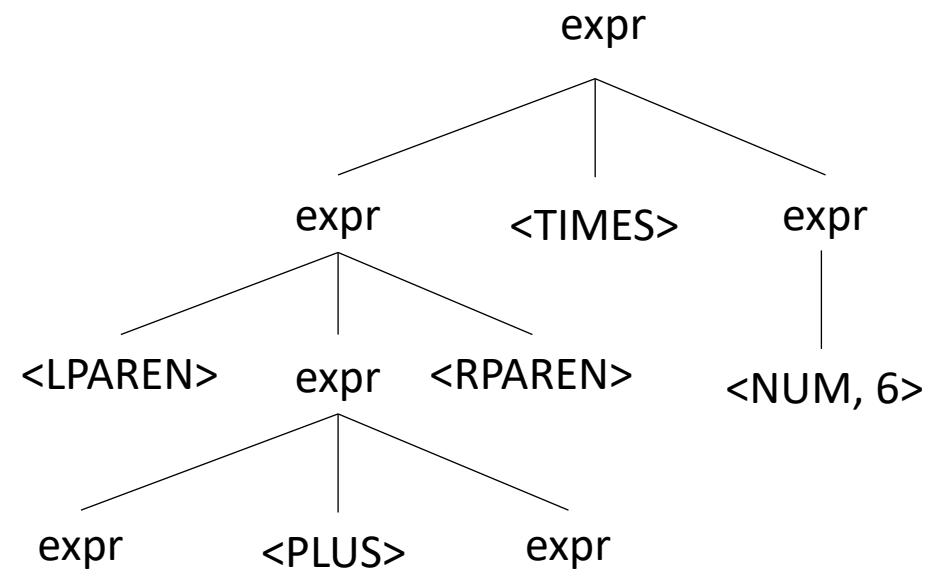
```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```



Parse trees examples

input: (1+5)*6

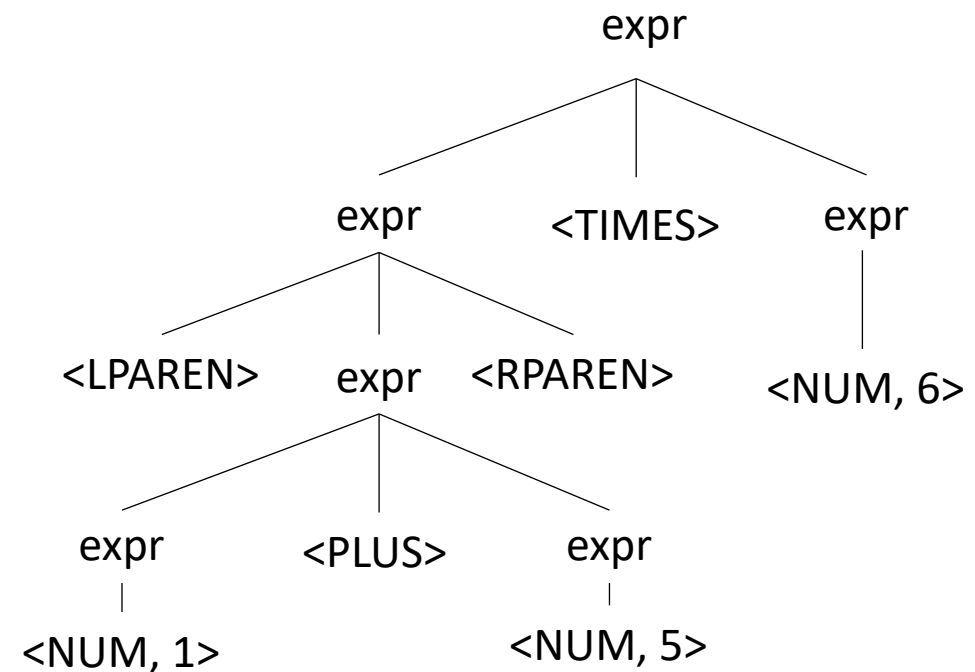
```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```



Parse trees examples

input: (1+5)*6

```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

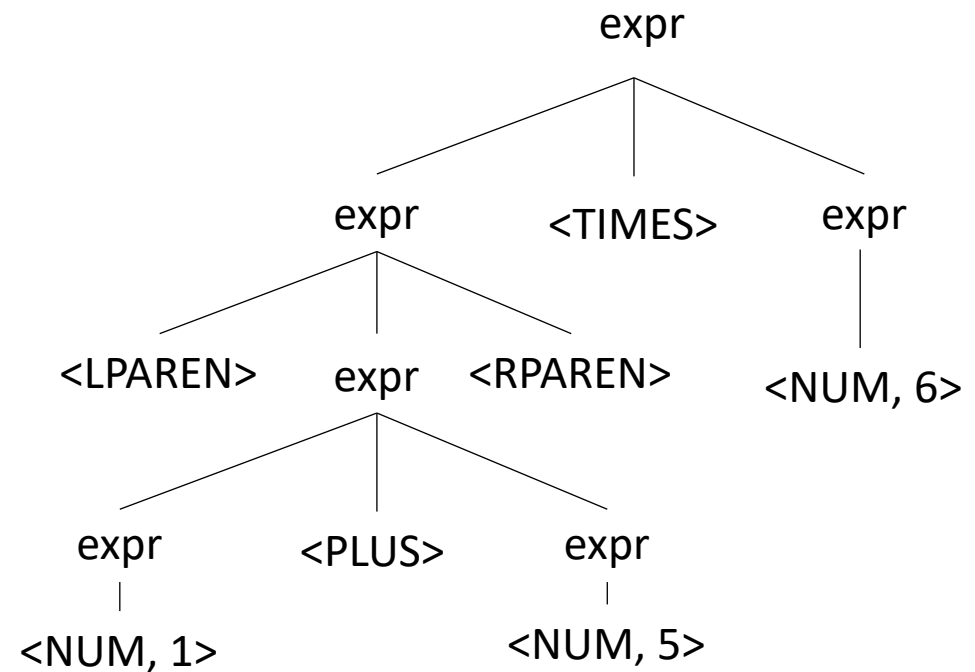


Parse trees examples

Does this parse tree capture the structure we want?

input: (1+5)*6

```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```



Parse trees

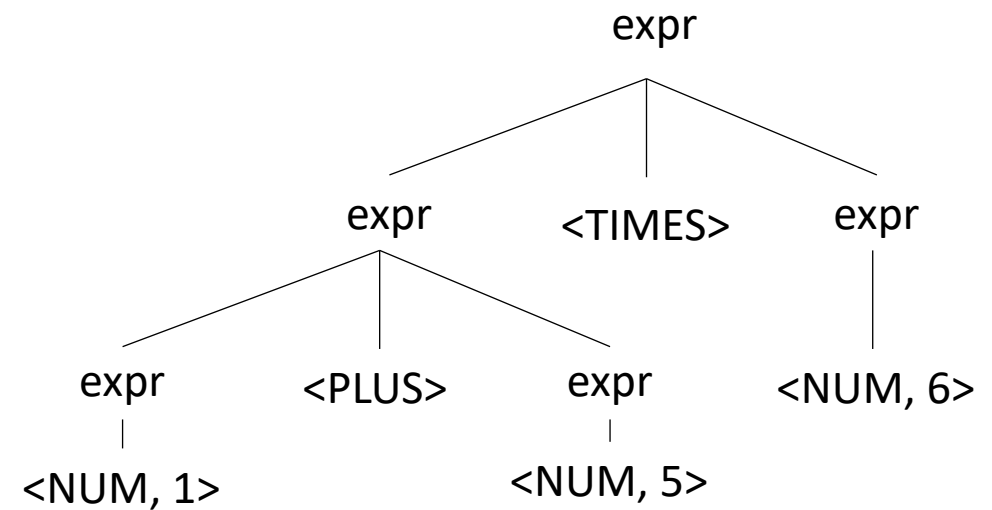
- How about: 1 + 5 * 6

```
expr ::= NUM  
      | expr PLUS expr  
      | expr TIMES expr  
      | LPAREN expr RPAREN
```

Parse trees

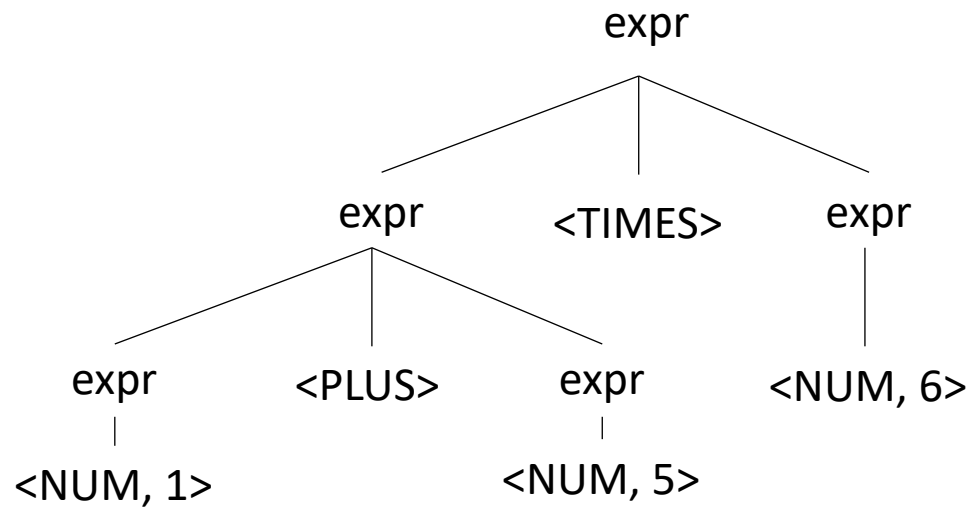
- How about: 1 + 5 * 6

```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```



Ambiguous Grammars

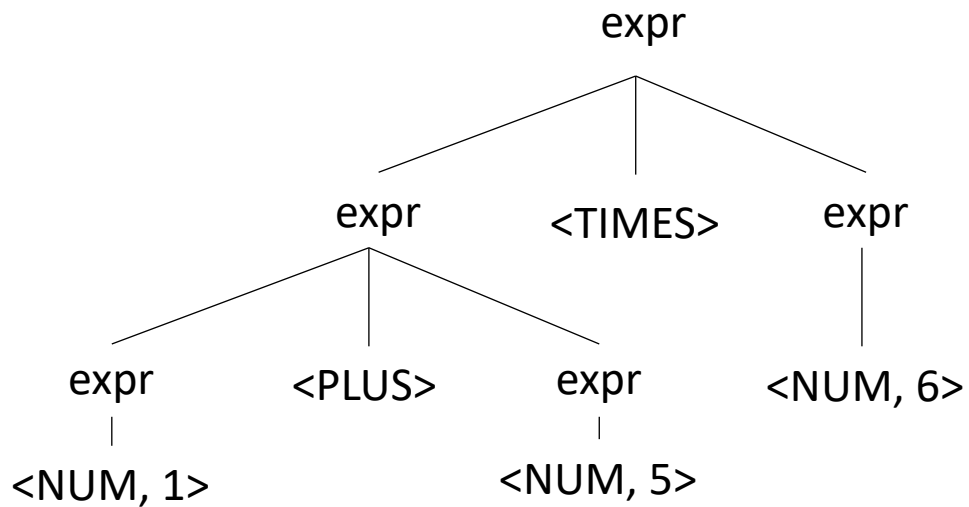
- input: 1 + 5 * 6



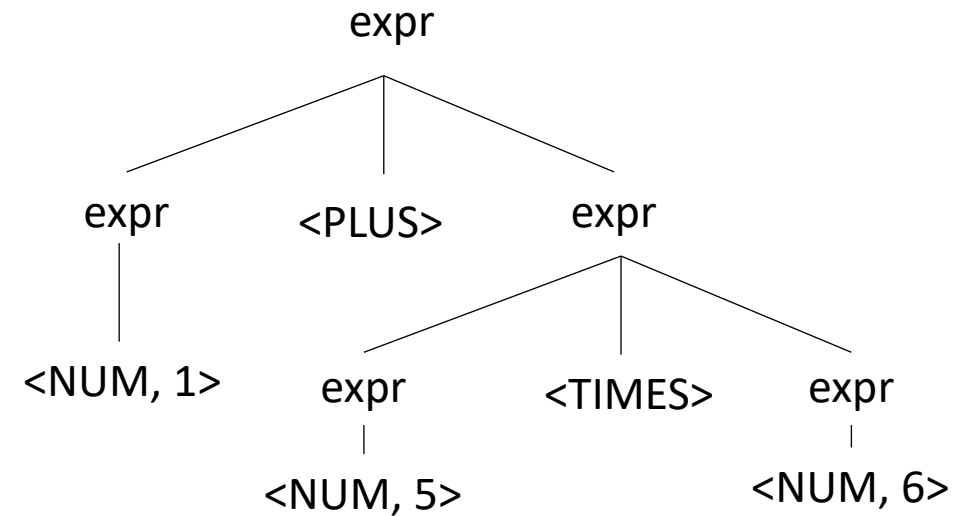
```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```

Ambiguous Grammars

- input: 1 + 5 * 6



```
expr ::= NUM
      | expr PLUS expr
      | expr TIMES expr
      | LPAREN expr RPAREN
```



Avoiding Ambiguity

- How to avoid ambiguity related to precedence?
- Define precedence into the grammar:
 - Ambiguity comes from conflicts. Explicitly define how to deal with conflicts by indicating that:
 - * has higher precedence than +
- Some parser generators support this,
 - e.g. YACC(C), Bison (C), Antlr (Java), PLY(Python)

Avoiding Ambiguity

- How to avoid ambiguity related to precedence?
- **Second way:** add new production rules
 - One non-terminal for each level of precedence
 - lowest precedence at the top
 - highest precedence at the bottom
- Lets try with expressions and the following:
+ * ()

Avoiding Ambiguity

Second way: new production rules

- One non-terminal for each level of precedence
- lowest precedence at the top
- highest precedence at the bottom

Operator	Name	Productions
+	expr	: expr PLUS expr term
*	term	: term TIMES term factor
()	factor	: LPAREN expr RPAREN NUM

Precedence
increases going down



Now lets create a parse tree

input: $1+5*6$

Operator	Name	Productions
+	expr	: expr PLUS expr term
*	term	: term TIMES term factor
()	factor	: LPAREN expr RPAREN NUM

Now lets create a parse tree

input: $1+5*6$

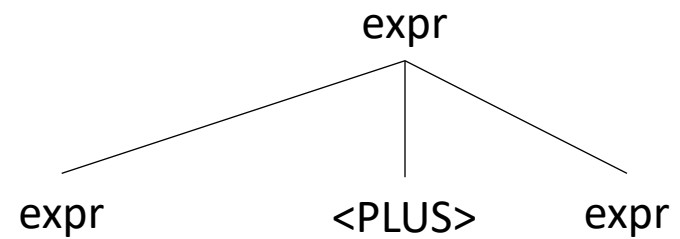
expr

Operator	Name	Productions
+	expr	: expr PLUS expr term
*	term	: term TIMES term factor
()	factor	: LPAREN expr RPAREN NUM

Now lets create a parse tree

input: 1+5*6

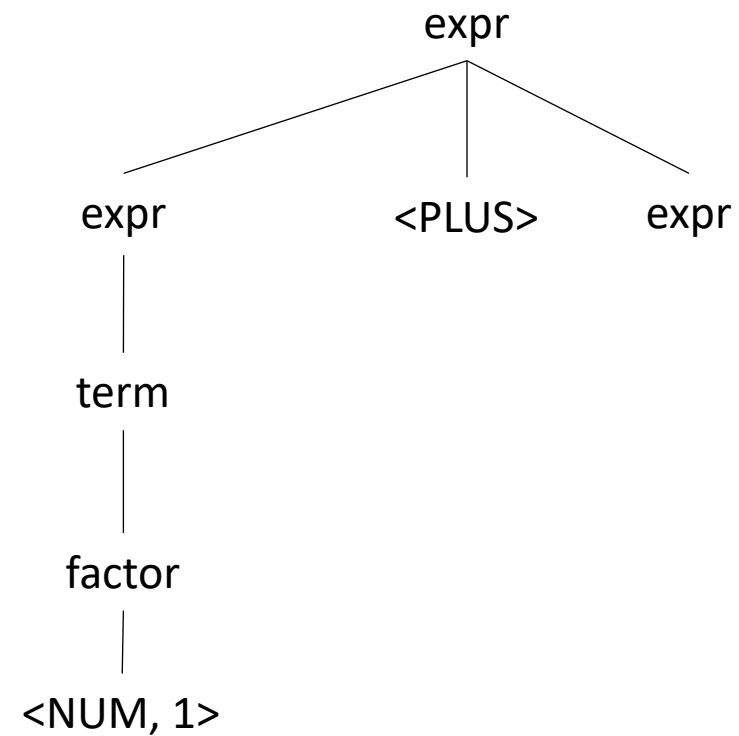
Operator	Name	Productions
+	expr	: expr PLUS expr term
*	term	: term TIMES term factor
()	factor	: LPAREN expr RPAREN NUM



Now lets create a parse tree

input: 1+5*6

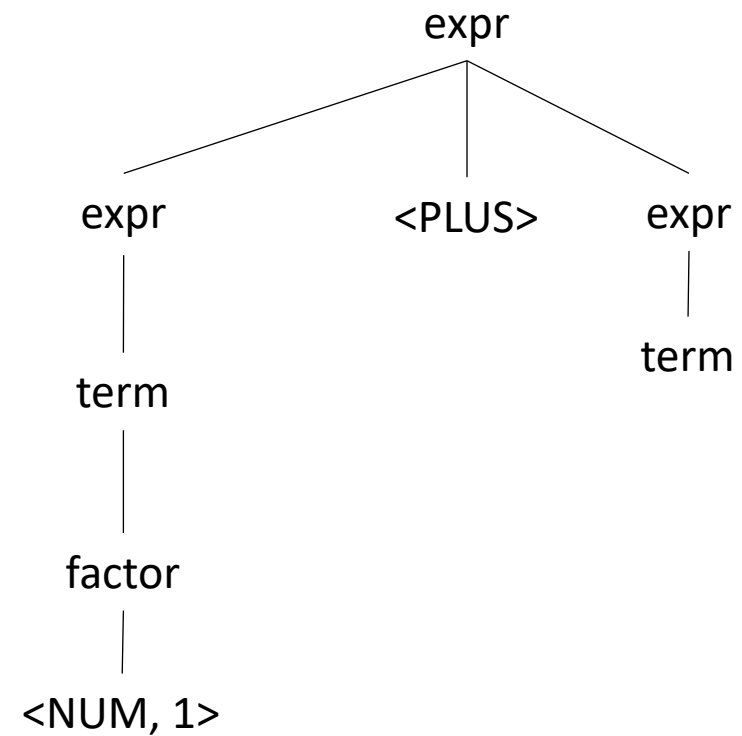
Operator	Name	Productions
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*	term	: term TIMES term factor
()	factor	: LPAREN expr RPAREN NUM



Now lets create a parse tree

input: 1+5*6

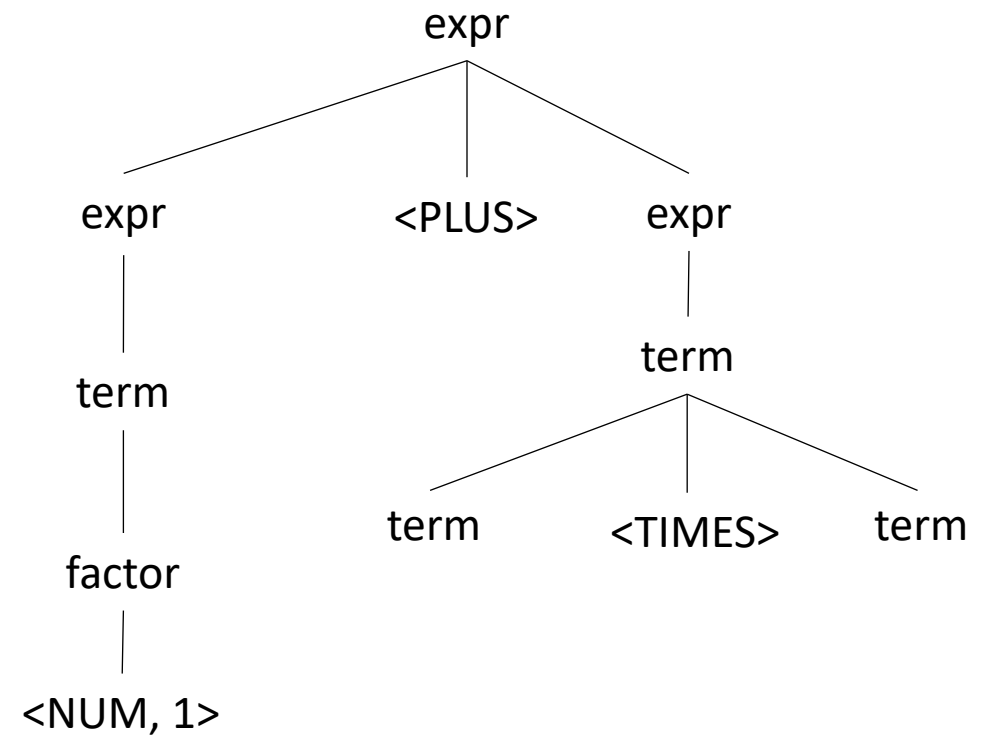
Operator	Name	Productions
+	expr	: expr PLUS expr term
*	term	: term TIMES term factor
()	factor	: LPAREN expr RPAREN NUM



Now lets create a parse tree

input: 1+5*6

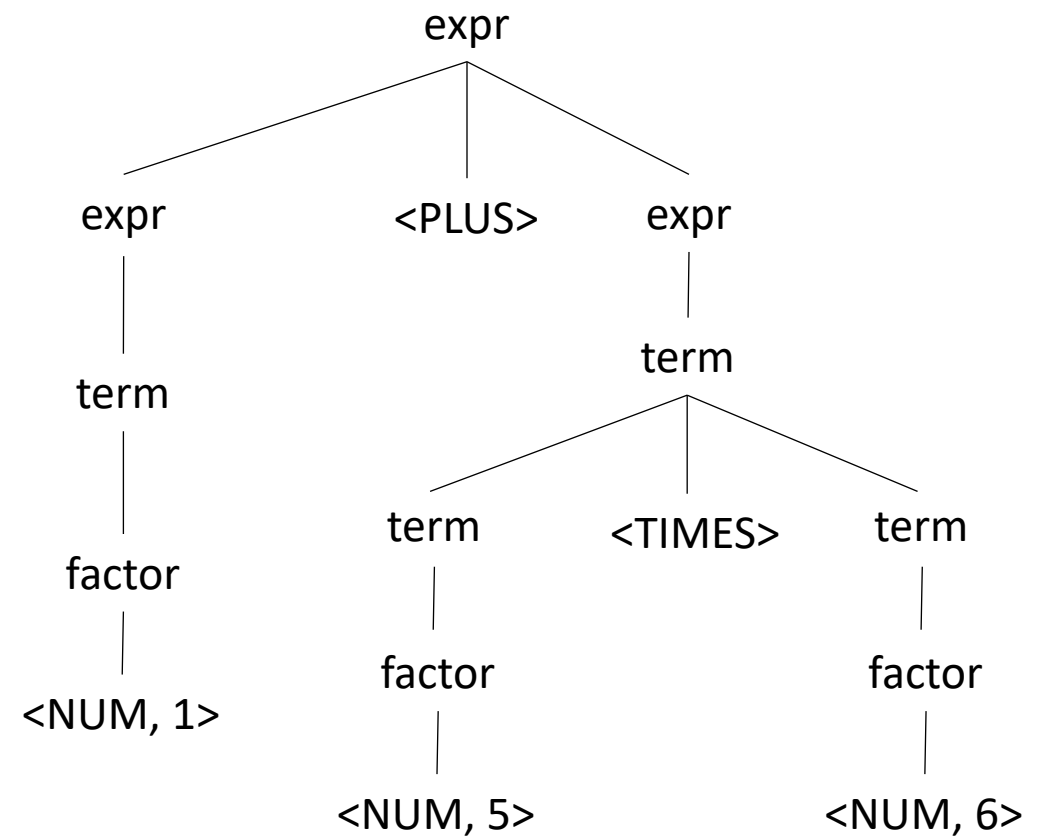
Operator	Name	Productions
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()	factor	: LPAREN expr RPAREN NUM



Now lets create a parse tree

input: 1+5*6

Operator	Name	Productions
+	expr	: expr PLUS expr term
*	term	: term TIMES term factor
()	factor	: LPAREN expr RPAREN NUM



Parsing Regular Expressions

(and considering precedence)

Let's try it for an RE language, $\{ | . * () \}$

- Assume $.$ is a concatenation operator
- Terminals are in upper-case

Tokens (Terminals)

PIPE = $|$

DOT = $.$

STAR = $*$

LPAR = $($

RPAR = $)$

CHAR = $[^a-zA-Z]$

Operator	Name (LHS)	Productions (RHS)

Possible names for NT (Non-Terminals):

choice

concat

starred

unit

Parsing Regular Expressions

(and considering precedence)

Let's try it for an RE language, $\{| \cdot * ()\}$

- Assume $\cdot$ is a concatenation operator
- Terminals are in upper-case

Operator	Name (LHS)	Productions (RHS)
	choice	choice PIPE choice concat
.	concat	concat DOT concat star
*	starred	starred STAR unit
()	unit	LPAR choice RPAR CHAR

Parsing Regular Expressions

(and considering precedence)

Let's try it for an RE language, $\{| \cdot * ()\}$

- Assume $\cdot$ is a concatenation operator
- Terminals are in upper-case

Operator	Name	Productions
	choice	: choice PIPE choice concat
.	concat	: concat DOT concat starred
*	starred	: starred STAR unit
()	unit	: LPAREN choice RPAREN CHAR

Parsing Regular Expressions

(and considering precedence)

Let's try it for an RE language, $\{| \cdot * ()\}$

- Assume $\cdot$ is a concatenation operator
- Terminals are in upper-case

input: $a.b \mid c^*$

Operator	Name	Productions
	choice	: choice PIPE choice concat
.	concat	: concat DOT concat starred
*	starred	: starred STAR unit
()	unit	: LPAREN choice RPAREN CHAR

Parsing Regular Expressions

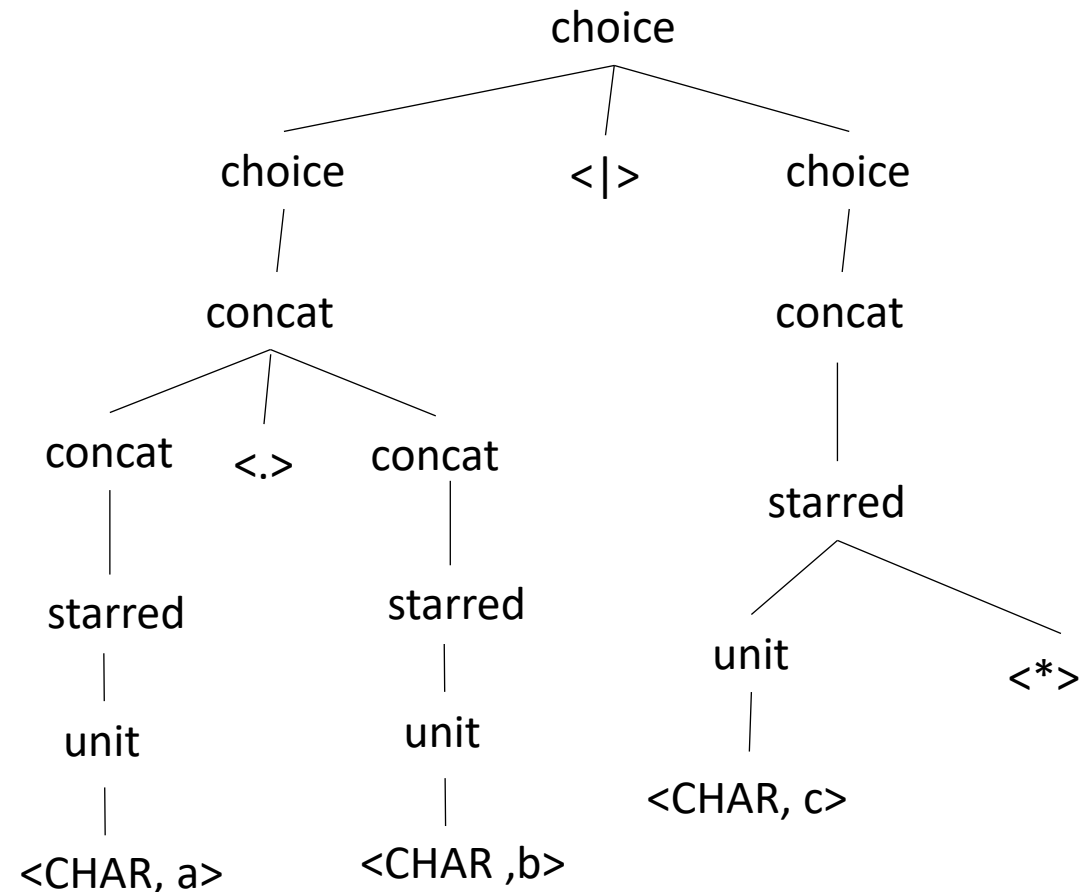
(and considering precedence)

Let's try it for an RE language, $\{ | . * () \}$

- Assume `.` is a concatenation operator
- Terminals are in upper-case

Operator	Name	Productions
	choice	: choice PIPE choice concat
.	concat	: concat DOT concat starred
*	starred	: starred STAR unit
()	unit	: LPAREN choice RPAREN CHAR

input: `a.b | c*`



How many levels of precedence does C have?

- https://en.cppreference.com/w/c/language/operator_precedence

Have we removed all ambiguity?

Let's make some more parse trees

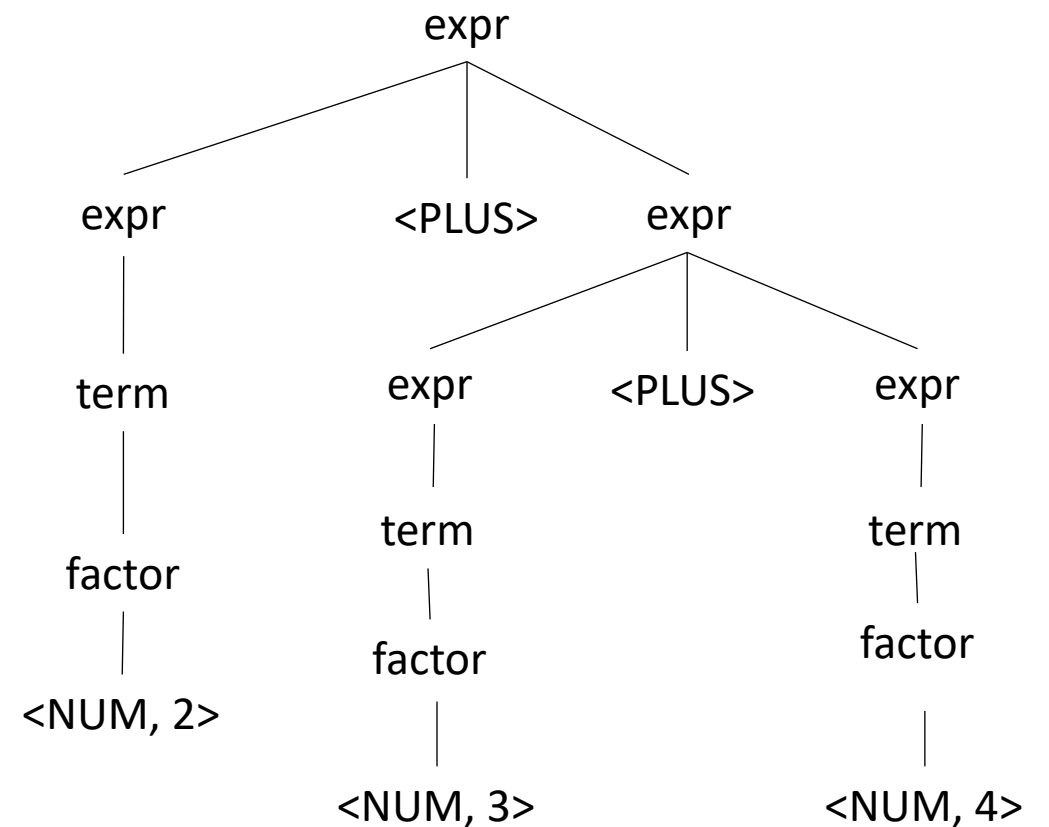
input: 2+3+4

Operator	Name	Productions
+	expr	: expr PLUS expr term
*	term	: term TIMES term factor
()	factor	: LP expr RP NUM

Let's make some more parse trees

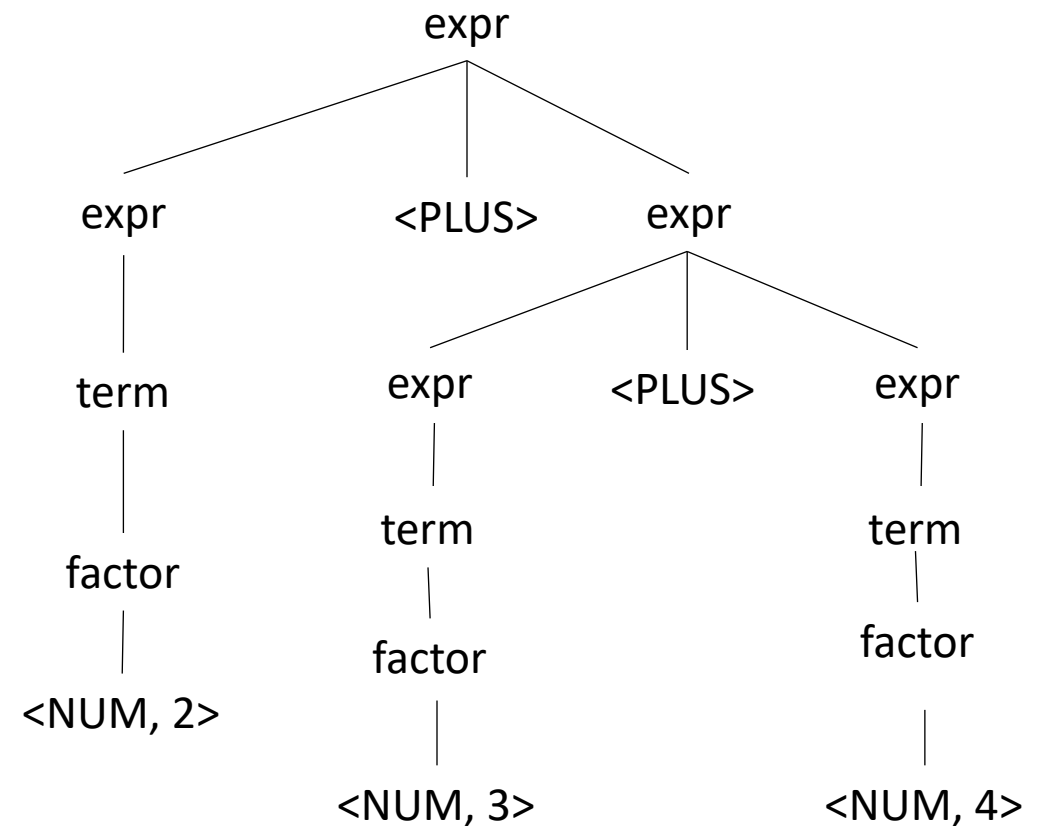
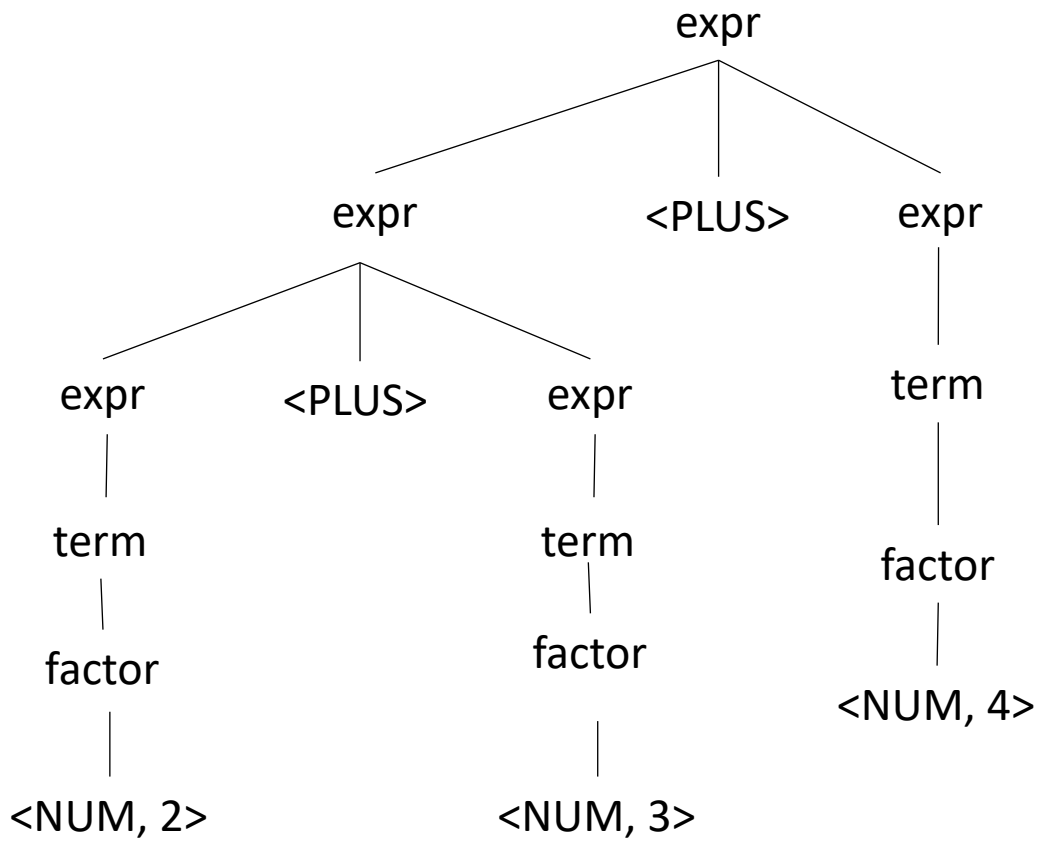
input: 2+3+4

Operator	Name	Productions
+	expr	: expr PLUS expr term
*	term	: term TIMES term factor
()	factor	: LP expr RP NUM



This is ambiguous, is it an issue?

input: 2+3+4

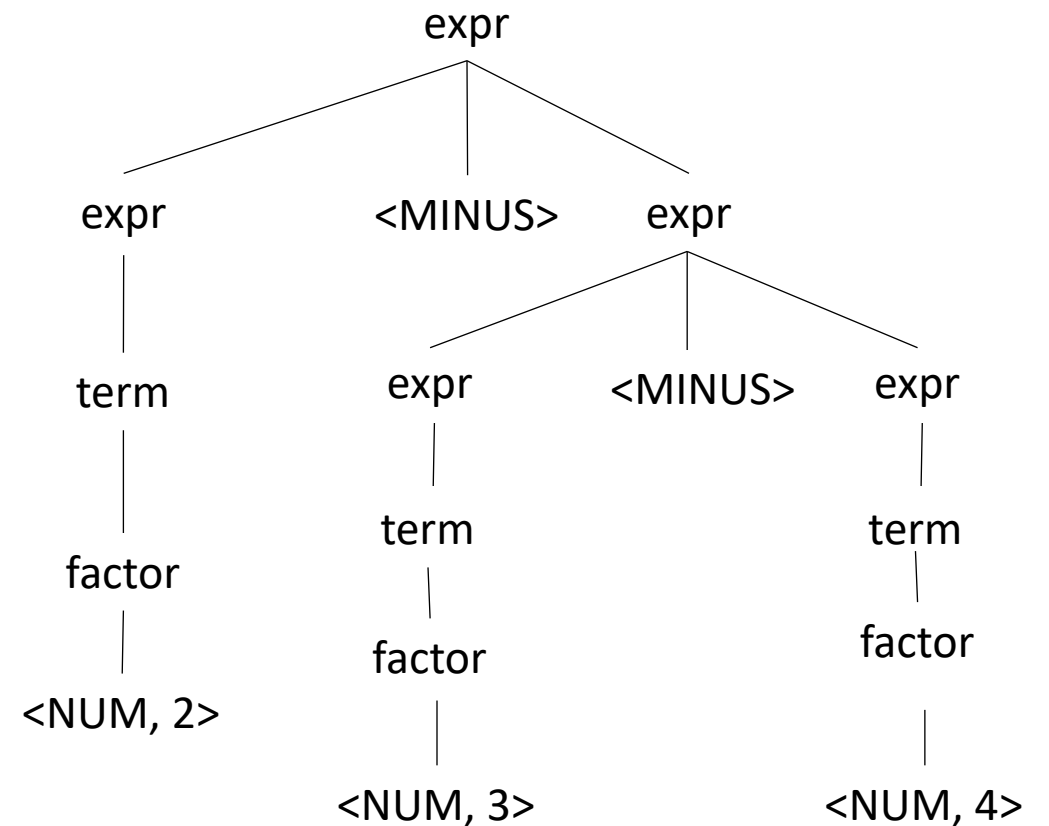
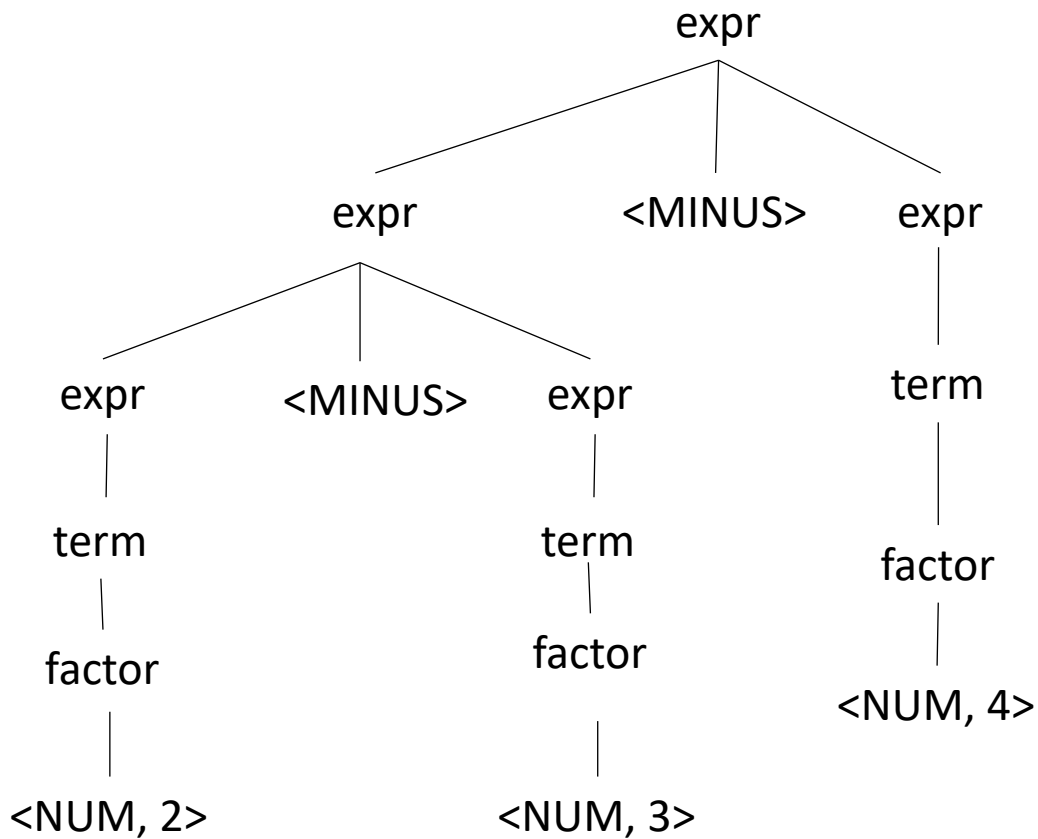


What about for a different operator?

input: 2-3-4

What about for a different operator?

input: 2-3-4



Which one is right?

Associativity

Describes the order in which apply the same operator

Sometimes it doesn't matter:

- When?

Associativity

Describes the order in which apply the same operator

Sometimes it doesn't matter:

- Integer arithmetic
- Integer multiplication

*These operators
are said to be associative*

Good test:

- $((a \text{ OP } b) \text{ OP } c) == (a \text{ OP } (b \text{ OP } c))$

What about floating-point arithmetic?

Associativity

If an operator is not associative then we define

- left to right (left-associative)
 - $2-3-4$ is evaluated as $((2-3) - 4)$
 - What other operators are left-associative
- right-to-left (right-associative)
 - Any operators you can think of?

Associativity

If an operator is not associative then we define

- left to right (left-associative)
 - $2-3-4$ is evaluated as $((2-3) - 4)$
 - What other operators are left-associative
- right-to-left (right-associative)
 - Assignment, power operator

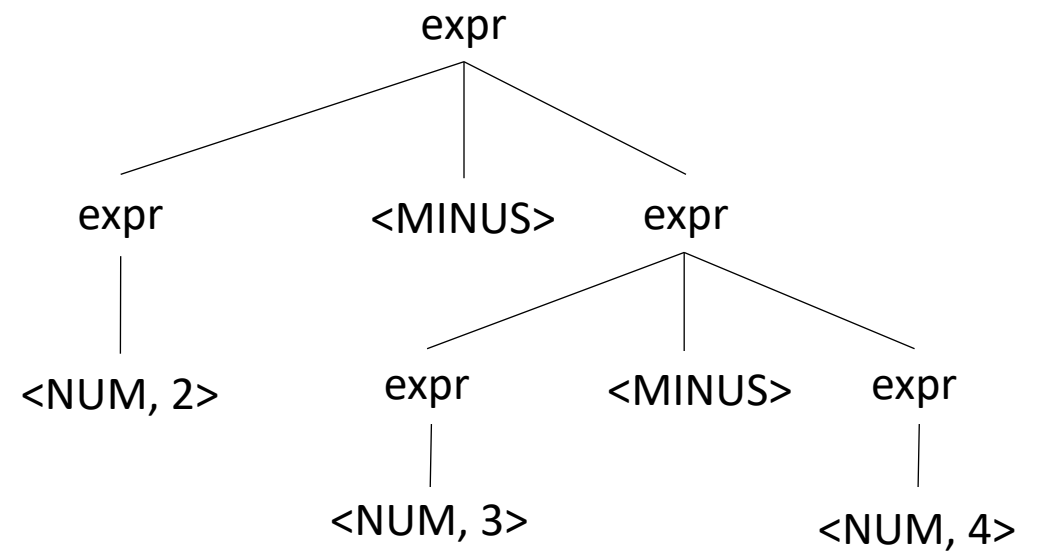
How to encode associativity?

- Like precedence, some tools (e.g. YACC) allow associativity specification through keywords:
 - “+”: left, “^”: right
- Like precedence, we can also encode it into the production rules

Ambiguous associativity for a single operator

input: 2-3-4

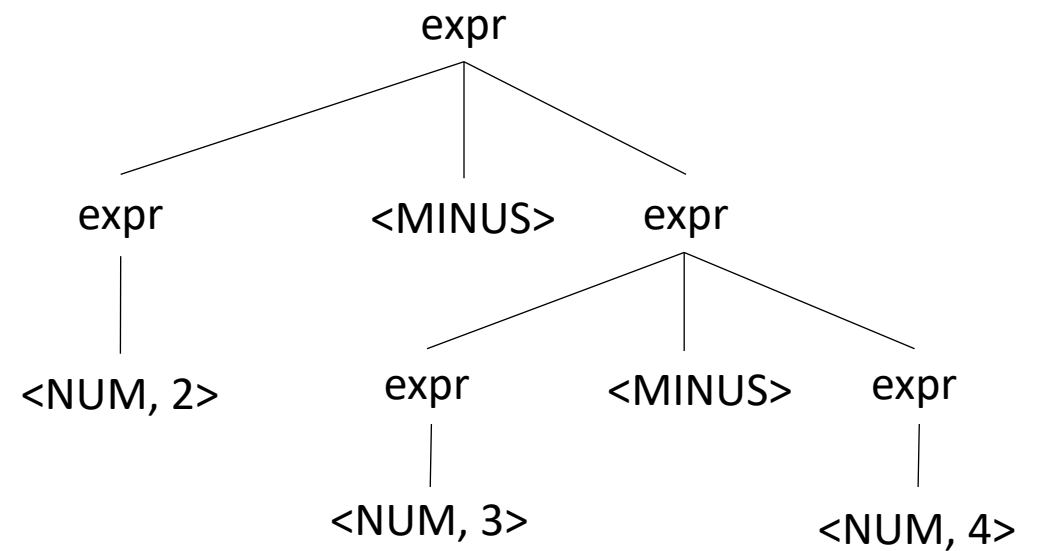
Operator	Name	Productions
-	expr	: expr MINUS expr NUM



Associativity for a single operator

input: 2-3-4

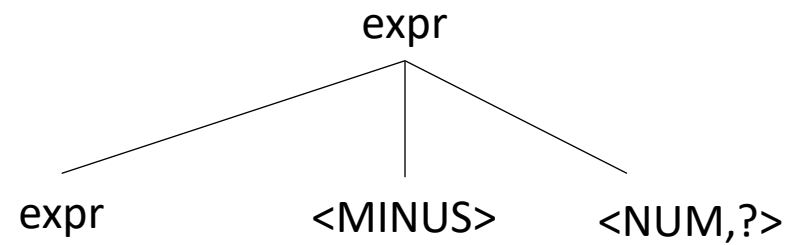
Operator	Name	Productions
-	expr	: expr MINUS NUM NUM



No longer allowed

Associativity for a single operator

input: 2-3-4

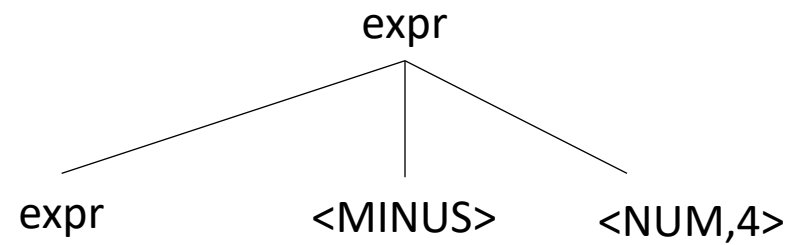


Operator	Name	Productions
-	expr	: expr MINUS NUM NUM

Lets start over

Associativity for a single operator

input: 2-3-4

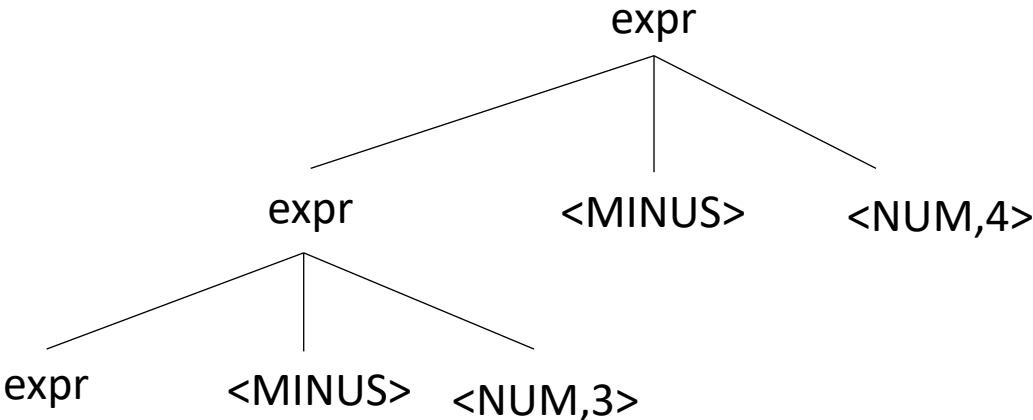


Operator	Name	Productions
-	expr	: expr MINUS NUM NUM

Associativity for a single operator

input: 2-3-4

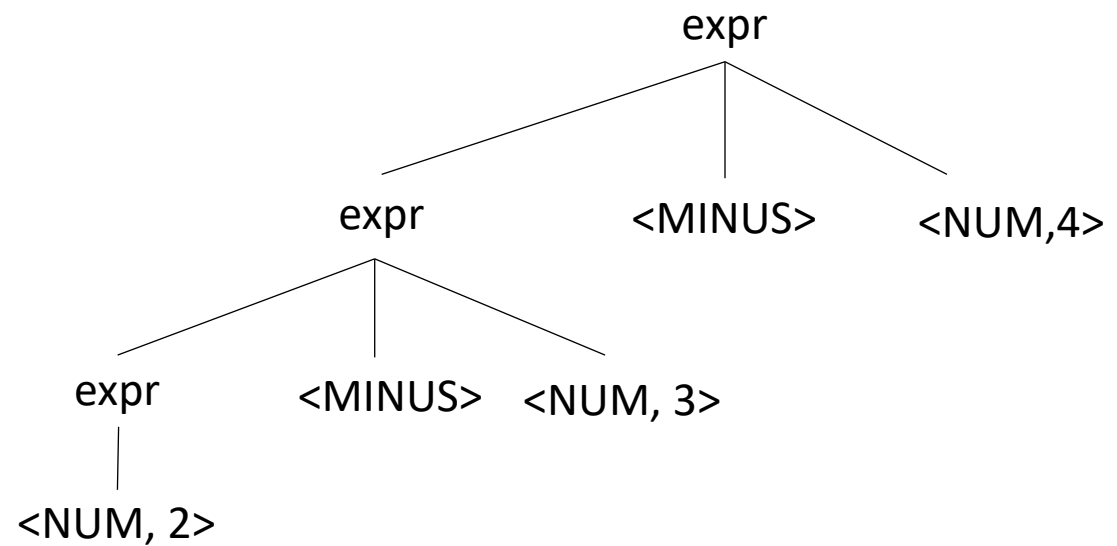
Operator	Name	Productions
-	expr	: expr MINUS NUM NUM



Associativity for a single operator

input: 2-3-4

Operator	Name	Productions
-	expr	: expr MINUS NUM NUM



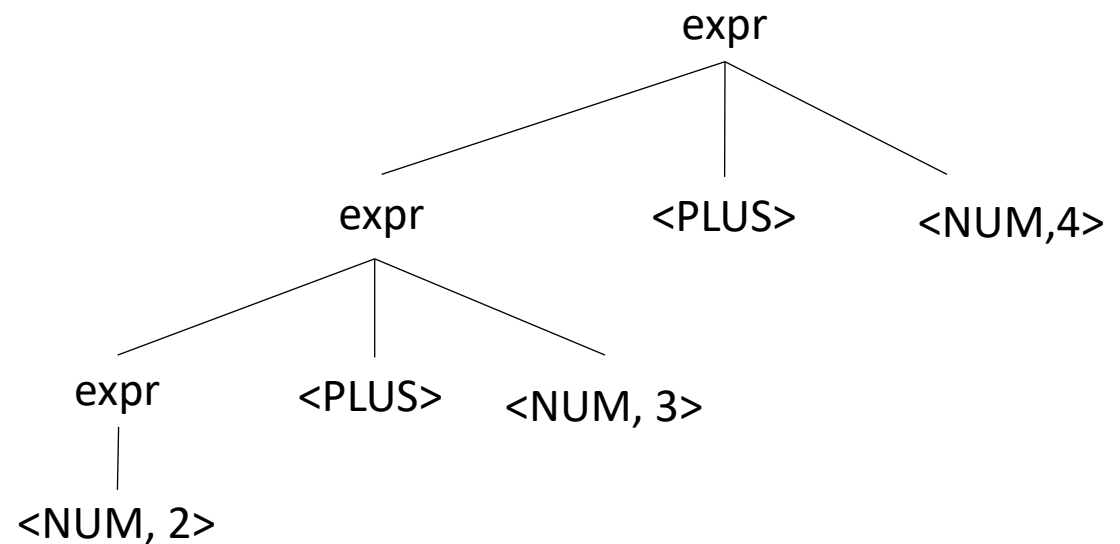
Left associative:
No longer ambiguous

Should you have associativity when its not required?

Benefits?
Drawbacks?

Operator	Name	Productions
+	expr	: expr PLUS NUM NUM

input: 2+3+4



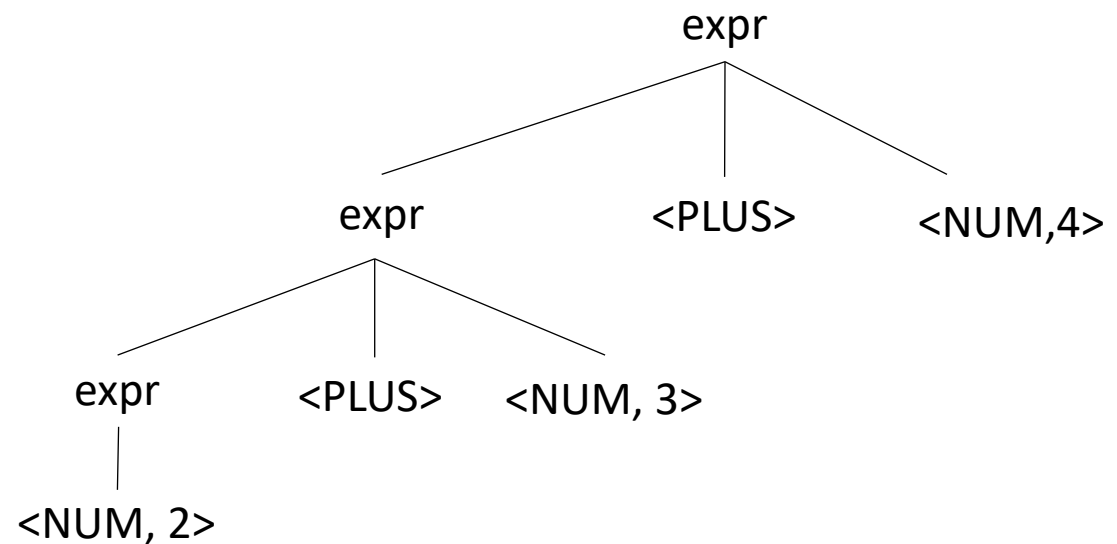
Should you have associativity when its not required?

Benefits?

Drawbacks?

Operator	Name	Productions
+	expr	: expr PLUS NUM NUM

input: 2+3+4



Good design principle to avoid ambiguous grammars, even when strictly not required too.

Helps with debugging, etc. etc.

Many tools will warn if it detects ambiguity

Let's make a richer expression grammar

*Let's do operators $[+, *, -, /, ^]$ and $()$*

Operator	Name	Productions

Tokens :

NUM = "[0-9]+"

PLUS = "\\+"

TIMES = "*"

LP = "\\("

RP = "\\)"

MINUS = "\\-"

DIV = "\\/"

CARROT = "\\^"

Possible names for NT (non-terminals):

expr
term
pow
factor

Let's make a richer expression grammar

Operator	Name	Productions
+, -	expr	: expr PLUS term expr MINUS term term
*, /	term	: term TIMES pow term DIV pow pow
^	pow	: factor CARROT pow factor
()	factor	: LPAR expr RPAR NUM

**Note the recursion from pow makes raising to a power right to left associative,
As opposed to PLUS, MINUS, TIMES DIV
Which are left-to-right associative.**

*Let's do operators $[+, *, -, /, ^]$ and $()$*

Tokens :

NUM = "[0-9]+"
PLUS = "\+"

TIMES = "*"
LP = "\ ("
RP = "\)"

MINUS = "\-"
DIV = "\/"

CARROT = "\^"

What associativities does C have?

- https://en.cppreference.com/w/c/language/operator_precedence

Next time: algorithms for syntactic analysis

- Top down parsing
 - oracle parsing
 - removing left recursion
 - constructing lookahead sets